AWS Security Groups Lesson

Skills you will learn

Cloud Computing: Compute & Networking

What Are Security Groups?

In Amazon Web Services (AWS), a **Security Group** acts as a virtual firewall for your EC2 instances. Security Groups control inbound and outbound traffic at the instance level. They are stateful, which means if you allow incoming traffic on a particular port, the corresponding outgoing response is automatically allowed, and vice versa. This simplifies managing access rules.

Security Groups are essential to:

- **Restrict Access**: Only allow necessary ports and protocols.
- **Secure Your Instances**: Prevent unwanted inbound connections.
- **Simplify Management**: Changes to a Security Group apply instantly to all instances associated with it.

Key Characteristics

- **Stateful** If an inbound rule allows traffic from a specific IP on a given port, the outbound traffic response to that IP and port is automatically allowed, even if there's no specific outbound rule for it.
- **Default Behavior**
  - **Inbound**: Denied by default (no inbound traffic is allowed unless specified).
  - **Outbound**: Allowed by default (all outbound traffic is allowed unless explicitly restricted).
- **Rules**
  - **Inbound Rules**: Specify which incoming connections are permitted (e.g., HTTP on port 80, SSH on port 22).
  - **Outbound Rules**: Specify which outgoing connections are permitted.
- **Attachments**

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- You can attach multiple Security Groups to an EC2 instance.
- The combination of all inbound rules across all attached Security Groups will be effective.

- **Rule Priorities** Unlike Network ACLs, there is no concept of rule numbering or priority in Security Groups. If traffic matches any allow rule, it is allowed. If it doesn't match any rule, it's denied.

Creating and Managing Security Groups

Step 1: Create a New Security Group

- In the AWS Management Console, navigate to **EC2** → **Security Groups** (under Network & Security).
- Click on **Create Security Group**.
- Provide a **Name** and **Description**.
- Choose the **VPC** where it should reside.
- (Optional) Add inbound/outbound rules right away or wait to add them later.

Step 2: Add Inbound Rules

When adding an inbound rule, specify:

- **Type** (e.g., SSH, HTTP, HTTPS, or custom TCP/UDP).
- **Port** (e.g., 22 for SSH, 80 for HTTP).
- **Source** (e.g., `0.0.0.0/0` for all IPv4 addresses or a specific IP range).

Step 3: Add Outbound Rules

- By default, outbound traffic is open. If needed, add or remove outbound rules to restrict traffic.

Step 4: Associate the Security Group with an EC2 Instance

- Either at instance launch or from the instance settings after launch (**Actions** → **Security** → **Change security groups**), attach your newly created Security Group to the EC2 instance.

Best Practices

1. **Least Privilege**: Only open the ports that you really need. Do not leave ports open to the world (`0.0.0.0/0`) if not necessary.
2. **Use Descriptive Names**: Name your Security Groups so it's clear what services/ports they allow.
3. **Regular Audits**: Periodically review inbound and outbound rules to ensure they match your requirements.
4. **Separate Security Groups by Function**: For example, one for web servers, one for database servers, and so on, to keep rules organized and easily manageable.

Hands-On Exercise

In this exercise, you will launch an EC2 instance, create and configure a Security Group, install Nginx, and verify that Security Group rules are working properly.

1. Launch an EC2 Instance

1. Go to the **EC2 Dashboard** in the AWS Management Console.
2. Click **Launch Instance**.
3. Select the Amazon Machine Image (AMI) (e.g., Amazon Linux 2).
4. Choose the instance type (e.g., `t2.micro`).
5. In the **Configure Security Group** step:
   - Create a new Security Group.
   - Name it something like `MyWebServerSG`.
   - Add **Inbound Rule** for:
     - **SSH** (port 22) to your IP (or `0.0.0.0/0` for testing, but remember best practices).
     - **HTTP** (port 80) from `0.0.0.0/0`.
6. Complete the launch process and wait for your instance to start.

2. Connect to Your Instance via SSH

1. Once the instance is running, note the **Public IPv4 address**.
2. Use your local terminal or a tool like PuTTY to SSH into the instance:

```
ssh -i /path/to/your_key.pem ec2-user@<YOUR_INSTANCE_PUBLIC_IP>
```

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3. You will be connected to your Amazon Linux instance.

3. Install Nginx

1. Update the package lists:

```
sudo yum update -y
```

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2. Install Nginx:

```
sudo amazon-linux-extras install nginx1 -y
```

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3. Start Nginx:

```
sudo systemctl start nginx
```

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4. Ensure Nginx starts on boot:

```
sudo systemctl enable nginx
```

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4. Test Remote Connection

1. Open a web browser.
2. Navigate to `http://<YOUR_INSTANCE_PUBLIC_IP>` (port 80).
3. You should see the default Nginx welcome page.

5. Block Port 80 in Your Security Group

1. Go to **EC2 Dashboard** → **Security Groups**.
2. Select `MyWebServerSG`.
3. Under the **Inbound** rules, remove or disable the rule allowing HTTP (port 80).
4. Wait a few seconds for the rule change to take effect.
5. Refresh the web page `http://<YOUR_INSTANCE_PUBLIC_IP>` in your browser.
   - The request should now fail to connect because port 80 traffic is no longer allowed.

6. Re-enable Port 80

1. Go back to **Security Groups** → **MyWebServerSG** → **Inbound** rules.
2. Add (or re-add) the **HTTP** rule on port 80 from `0.0.0.0/0`.
3. Save the rule, wait a few seconds, and refresh the web page.
   - You should see the Nginx welcome page again.

Summary

Security Groups are a critical component of AWS security. By configuring inbound and outbound rules properly, you control the flow of traffic to your instances. Remember to follow the principle of least privilege and only allow what's necessary. In this exercise, you saw how enabling and disabling a port in a Security Group directly affects the availability of services running on your EC2 instance.

Congratulations! You have now created and tested a Security Group on AWS, installed a web server (Nginx), and verified how Security Group rules can allow or block traffic. This knowledge is fundamental when deploying and securing services in the AWS environment.

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